

MISSION 1

The Questions

- Question 1

Consider the two dimensional autonomous system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

Suppose our domain is simply connected , Is the following statment correct?

$$\operatorname{div} V = 0 \iff A \text{ is a } \textit{Hamiltonian system}$$

- Question 2

Suppose we have the two dimensional autonomous system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

And the complex form of A is given by:

$$\tilde{A} : \dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

Supoose :

$$\begin{aligned} A &:= A_1 + iA_2 \\ B &:= B_1 + iB_2 \\ C &:= C_1 + iC_2 \\ \mu &:= (A_1, A_2, B_1, B_2, C_1, C_2) \end{aligned}$$

Find the condition of μ such that $\tilde{A}$ is a *Hamiltonian system*.

The Answers

- Answer of Question 1 :

YES , it is correct.

PROOF:

($\Leftarrow$) : We have the hypothesis that *A is a Hamiltonian system* that is $\exists H : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $H \in$

$C^2(\mathbb{R}^2)$ such that $\begin{cases} P(x, y) = H_y \\ Q(x, y) = -H_x \end{cases}$ Now , we have the definiton of $\operatorname{div} V$: *Definiton* : Suppose we

have the two dimensional autonomous system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

Then we define the $\operatorname{div} V := \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$

We want to show that $\operatorname{div} V = 0$ So $\operatorname{div} V \stackrel{\text{by definition}}{=} \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \stackrel{\text{hypothesis}}{=} H_{yx} + (-H_{xy}) \stackrel{H \text{ is in } C^2}{=} 0 \square$
 $(\Rightarrow) :$

We need to introduce some definitions first :

Definiton 1(1 - form) : We say that w is a 1 - form if $w = A(x, y)dx + B(x, y)dy$ where $A, B : \mathbb{R}^2 \rightarrow \mathbb{R}$ are fucntions.

Definiton 2(Exact form) : Given a 1 - form $w = A(x, y)dx + B(x, y)dy$ we say w is exact if $w = dH$ for some H Where $dH := \frac{\partial H}{\partial x}dx + \frac{\partial H}{\partial y}dy$

Definiton 3(Closed form) : Given a 1 - form $w = A(x, y)dx + B(x, y)dy$ we say that w is closed if $dw = 0$
 (where $dw := dA \wedge dx + dB \wedge dy = (A_x dx + A_y dy) \wedge dx + (B_x dx + B_y dy) \wedge dy = (B_x - A_y)dx \wedge dy$)

Theorem(Poincare lemma) : Given a 1 - form $w := A(x, y)dx + B(x, y)dy$ and the domain of A, B are simply connected. Then w is exact $\iff w$ is closed

Now back to our proof, We have the hypothesis that $\operatorname{div} V = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0$ We want to find H such that $H_y = P$ and $-H_x = Q$.

Since $dH = H_x dx + H_y dy = -Qdx + Pdy$

$$\text{So } \exists H \iff w := -Qdx + Pdy \text{ is exact} \quad (*)$$

Now compute dw :

$$w := -Qdx + Pdy \Rightarrow dw = (P_x + Q_y)dx \wedge dy \stackrel{\text{hypothesis}}{=} 0dx \wedge dy = 0$$

So w is closed

by *Poincare lemma* , we konw that w is exact , then by $(*)$ we konw that $\exists H$ That is , the system is *Hamiltonian*. $\square$

- Answer of Question 2 :

The condition is : $2A = -\bar{C}$ and B is free.

Consider the complex form of two dimensional autonomous system :

$$\dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z} \quad (*)$$

where

$$A := A_1 + iA_2$$

Now we try to change this complex form into the original form Supoose

$$z := x + iy$$

Then $(*)$ is equal to

$$\dot{z} := \dot{x} + i\dot{y} = i(x + iy) + A(x + iy)^2 + B(x - iy)^2 + C(x + iy)(x - iy)$$

$\Rightarrow$

$$\dot{z} = i(x + iy) + (A_1 + iA_2)(x^2 - y^2 + 2xyi) + (B_1 + iB_2)(x^2 - y^2 - 2xyi) + (C_1 + iC_2)(x^2 + y^2)$$

$\Rightarrow$

$$\dot{z} \stackrel{\text{after simplify}}{=} (-y + (A_1 + B_1)(x^2 - y^2) + (2B_2 - 2A_2)xy + C_1(x^2 + y^2)) \\ + i(x + (2A_1 - 2B_1)xy + (A_2 + B_2)(x^2 - y^2) + C_2(x^2 + y^2))$$

Now we have the original form :

$$\begin{cases} \dot{x} = -y + (A_1 + B_1)(x^2 - y^2) + (2B_2 - 2A_2)xy + C_1(x^2 + y^2) \\ \dot{y} = x + (2A_1 - 2B_1)xy + (A_2 + B_2)(x^2 - y^2) + C_2(x^2 + y^2) \end{cases}$$

Now compute the $\text{div}V = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = (2(A_1 + B_1)x + (2B_2 - 2A_2)y + 2C_1x) + ((2A_1 - 2B_1)x - 2(A_2 + B_2)y + 2C_2y)$

In Question 1 , We know that :

$$\text{div}V = 0 \iff A \text{ is a Hamiltonian system}$$

Let $\text{div}V = 0$,Then we have :

$$\text{div}V = (4A_1 + 2C_1)x + (-4A_2 + 2C_2)y = 0 \quad \forall x, y$$

So we have :

$$\begin{cases} 4A_1 + 2C_1 = 0 \\ -4A_2 + 2C_2 = 0 \end{cases}$$

$\Rightarrow$

$$\begin{cases} 2A_1 = -C_1 \\ 2A_2 = C_2 \end{cases}$$

That is :

$$\begin{aligned} 2A &= -\bar{C} \\ B &\text{ is free} \end{aligned}$$

□